

Feras Bouteraa

Third Year Mechanical Engineer

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Education

De Montfort University (Graduating 2026):

- Currently achieving 69% average. Participating in DMU's formula student team.
- Sole designer for the suspension and steering components of the 2027 competition race car, using Prog3d, CREO & FEA. Chosen to gain experience in a project, with the aim of learning and, the objective of winning the competition. Researching, designing & deciding how to succeed; balancing optimisation against cost, manufacturing & geometry limits.
- Collaborating with the chassis & upright designer and use Susprog3d to find optimal suspension nodes within specification. Using CREO & ANSYS to iteratively improve the design to be light weight and effective. Knowledge of suspension components, wheel alignment & suspension systems (e.g. Damping Constant, Camber, MacPherson etc.)

Recent Modules:

- Effectively designing a product; product design specification, morphological charts, Pugh matrix, materials selection & manufacturability. filament methods & real-world problems.
- Lab experiment and report of diesel engine emissions at different rpm; BMEP, BSFC, A/F. Euro emissions ratings history and future, advancements in emissions technologies.
- Fatigue analysis report of aluminium, steel & CFRP via in house experimentation. The material attributes required for high fatigue resistance; like that of Carbon Fiber. Etc.

Skills

- CREO: Designs using feature-based parametric modelling tools, understanding design-intent, ability to form design assemblies. Simplistic, efficient designs based around manufacturability and cost; a CNC machine can cut sharp edges on the outside, but not the inside of a material. Simple and effective designs are manufacturable and hit criteria.
- FEA (ANSYS): Evaluate structural safety, critical stress regions, topology optimisation iteratively improve design. Reasonable mesh size and refine mesh for time efficiency.
- Susprog3d: Input centre of mass, roll centre height and virtual swing arm length to retrieve ideal suspension geometry. Learning from a past student and current experienced supervisor. Experimenting with Formula Student Suspension geometry.
- 3D Printing & Design (ANSYS): topology optimisation for non-compliance in ANSYS, apply Voronoi infill to finger cast, Repair missing faces on Spaceclaim & convert to stl file.

Experience

- Caterpillar UK, Desford. Worked as high pressure hose cutter. Maintained safety and time management in a fast-paced environment around heavy-duty equipment.
- Amazon UK, DNG1. Delivery driver under KMI Logistics. Kept composure under stress of unhappy and abusive customers & improved time management from feedback.
- Impact Retail Ltd, Meridian S. Assembly of commercial product stands using hot glue. First paid work experience at 18, learnt how to work in a team with different people.